Cardiac spatial organization and CM maturation

Question: are ventricular CMs near proliferating or sprouting-associated ECs less mature, particularly at P0/P7?

Three principal lineages: cardiomyocytes, fibroblasts and endothelial cells; all other cardiac profiles remain in the background.

Maturation: measured mean log1p(CP10k) of the exact 21 Figure4b genes, with fixed-control, Myl2-withheld and boundary-core sensitivities.

EC proliferation: directly measured Mki67/Top2a/Cenpf nuclear RNA, multi-marker support and the separate RNA-inferred proliferation module.

Sprouting: exploratory RNA-inferred six-gene module; only Angpt2 of that signature is measured by Xenium. The missing-five component is tracked separately.

A new alignment sensitivity excludes all 21 CM maturation genes and all sprouting targets from EC predictors and their normalization.

Primary neighborhood: at least3 eligible ECs within 50µm and at least5 resolved cardiac neighbors;30/100µm are sensitivities.

Adjustment: CM capture/area, CM cycling/Slit2, EC capture/CM-marker fraction, EC subtype composition, local lineage density/composition, XY splines and vascular/endocardial proxies.

Spatial uncertainty:400µm block bootstrap;200/800µm sensitivities; buffered spatial prediction. No independent animal-level significance is claimed.

Confidence-qualified fibroblast and EC families are displayed without assigning unsupported fine states or treating gray profiles as absent.

One Xenium section per age; centroid proximity does not demonstrate contact, causation, active sprouting or completed cell division.

Findings: spatial organization and the proximity hypothesis

P0/P7: no convincing reduction in the overall 21-gene maturation score near cycling ECs.

Stage	Mean 21-gene score	Near cycling EC	Other CMs	Adjusted near−other	95% spatial-block interval
P0	2.532	7,012	10,731	+0.0056	[-0.0073, +0.0187]
P7	2.675	11,042	25,203	+0.0062	[-0.0009, +0.0125]
P14	2.923	730	27,214	+0.0035	[-0.0213, +0.0293]
P21	2.983	445	25,677	-0.0375	[-0.0659, -0.0029]

Near: CM centroid≤30μm from an eligible EC with≥2 distinct nuclear Mki67/Top2a/Cenpf genes. Comparison: other eligible CMs in the same section; both groups have≥3 eligible ECs within 50μm. Effects use log-expression score units.

Direct nuclear cycling, inferred proliferation and relative distance to cycling versus marker-negative ECs give no convincing overall immaturity association at P0/P7. Boundary-core and fixed-control score checks support that conclusion.

Early inferred sprouting has a tiny positive association with maturation (~0.02 outcome SD/exposure SD), not the predicted negative direction. Missing-gene recovery is weak; this is not a verified sprouting niche.

An exploratory P0 sarcomere-only component is modestly lower with greater local nuclear cycling (~−0.045 SD); the full panel and P7 do not show that pattern. It is not evidence for globally less mature CMs.

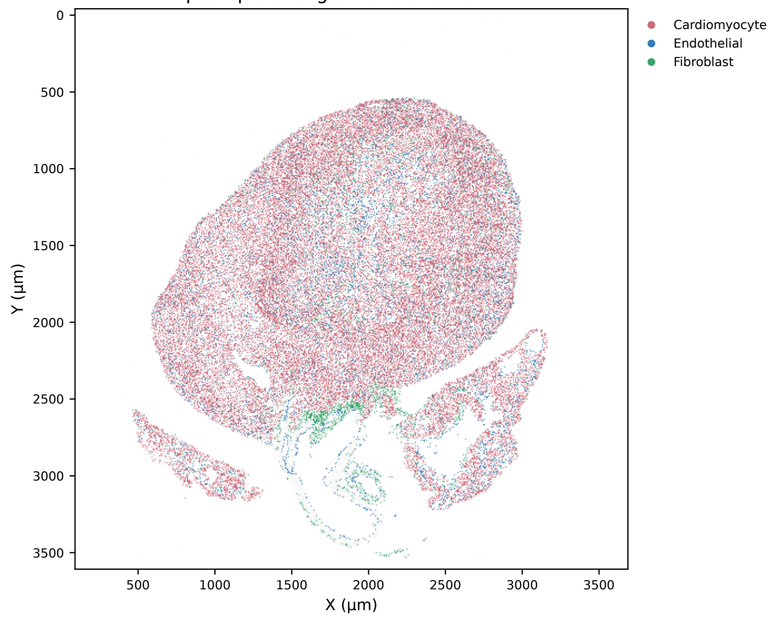
P21 has a small negative near-cycling contrast in native counts. It attenuates with CM boundary-core counts and is uncertain in the author-compatible subset; this is boundary-sensitive rather than a firm late niche.

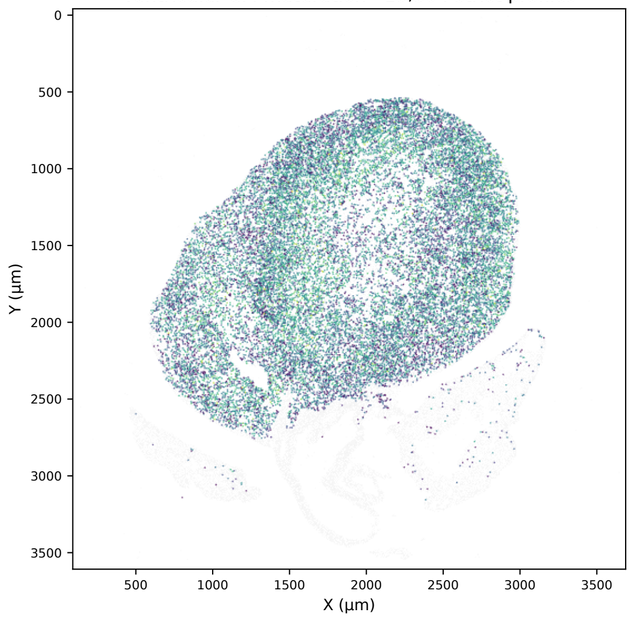
Spatially: capillary-like ECs are interspersed with CMs; endocardial-like ECs favor lining-associated territories; arterial-like ECs have many more smooth-muscle neighbors. Small venous/lymphatic candidates remain provisional.

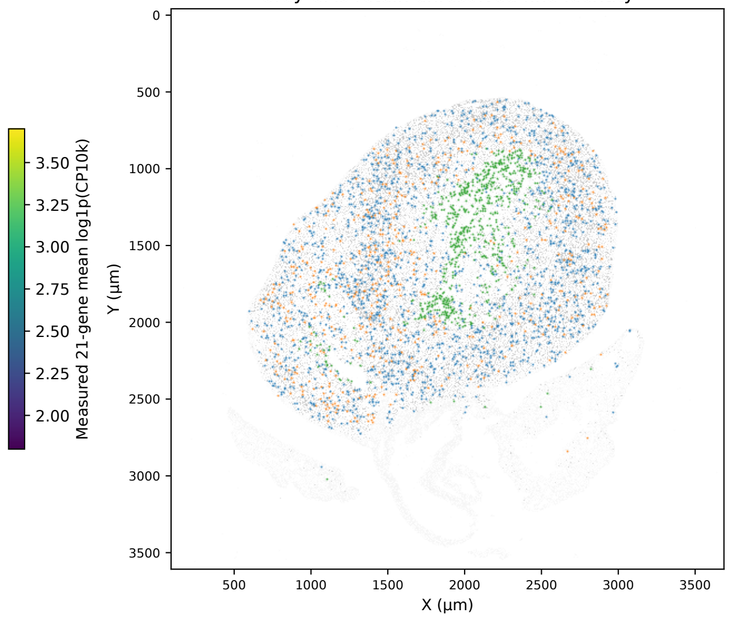
Early Postn fibroblasts occupy CM-containing interstitial territories. Later Col8a1 and Gfpt2 families have distinct stromal/vascular contexts; separate P21 RNA substates remain unresolved spatially. Conduction-program CM candidates lie closer to endocardial proxies.

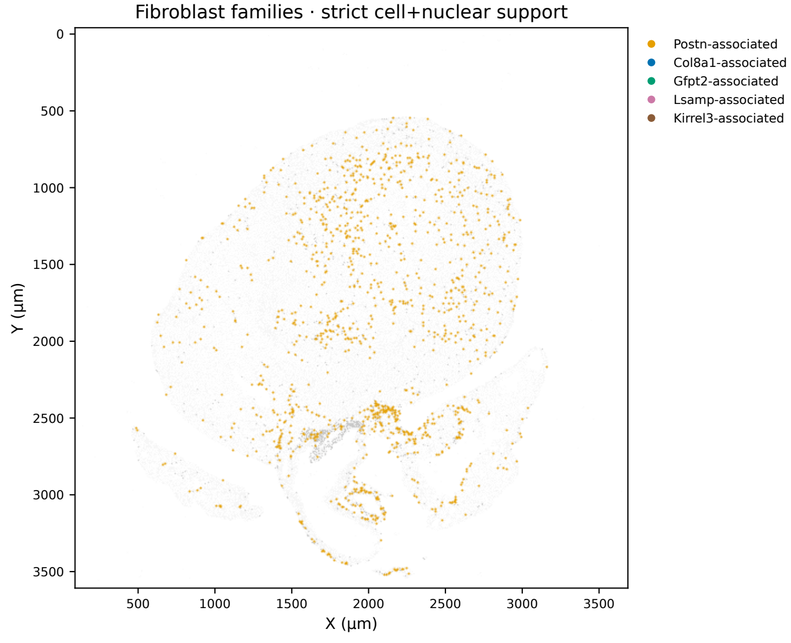
Across-age maturation increases while EC cycling-marker support decreases. That developmental co-trend does not establish a within-age local relationship or causality. One section/age limits generalization.

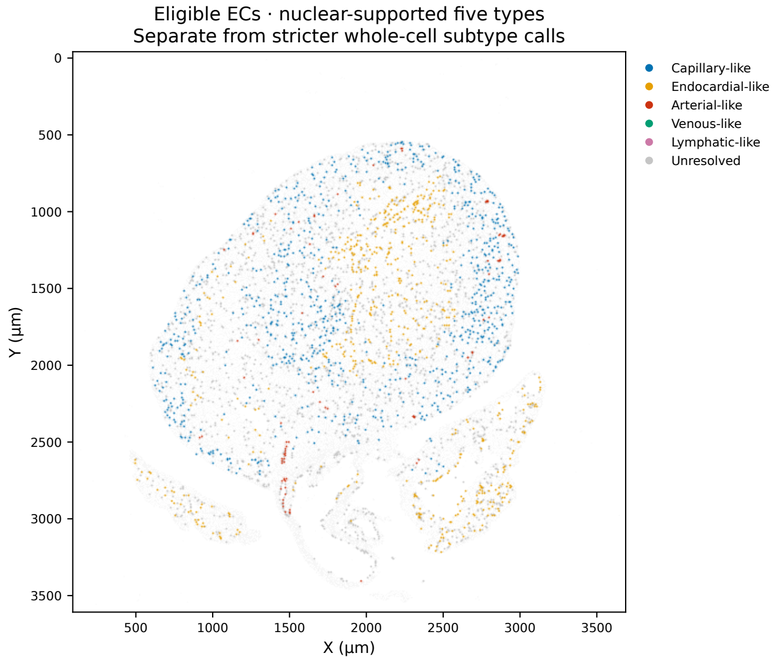
P0 · cardiac lineages, supported subtypes and ventricular maturation
Native section coordinates; gray fine identities are retained; confidence-selected coverage is incomplete

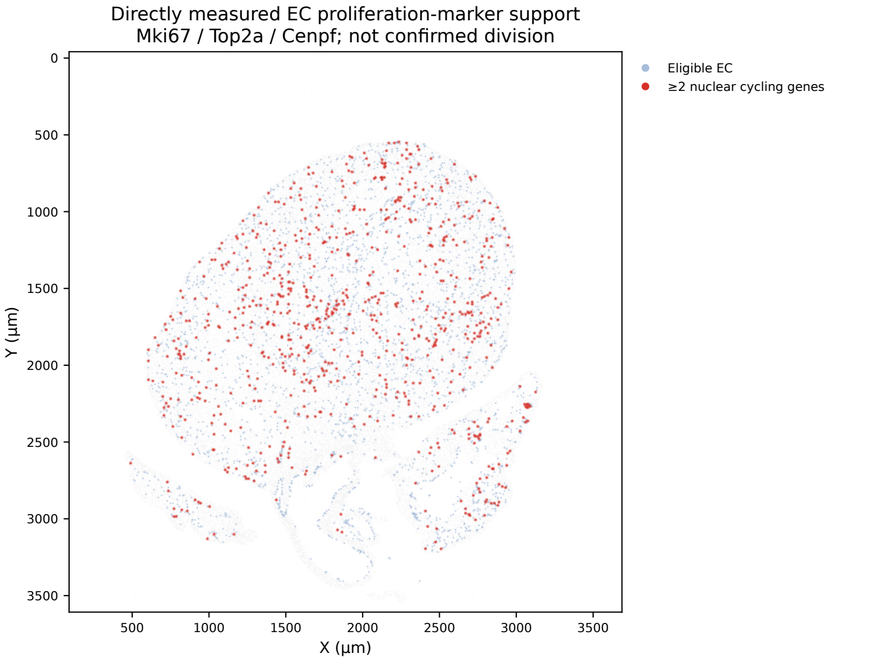




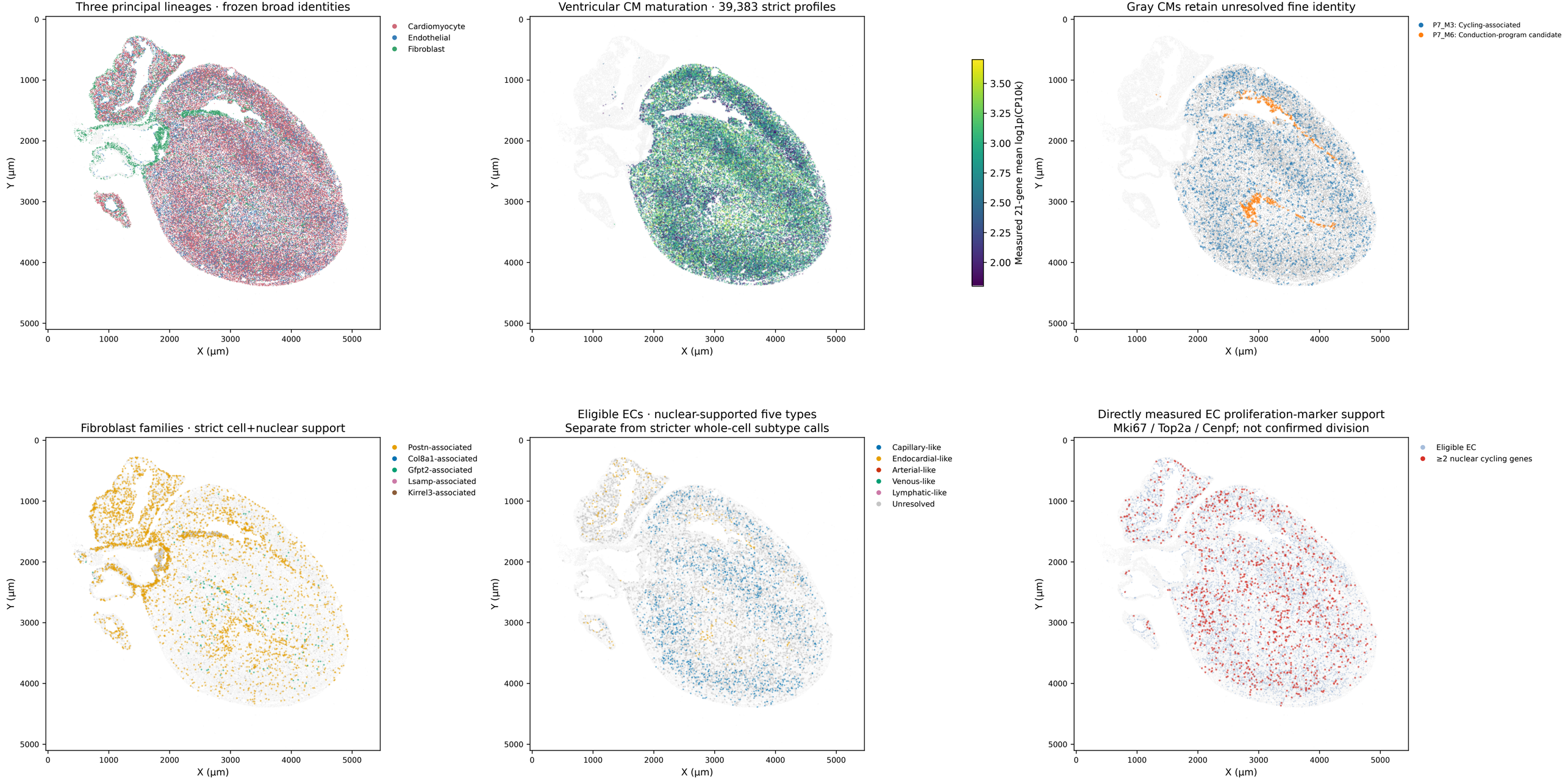






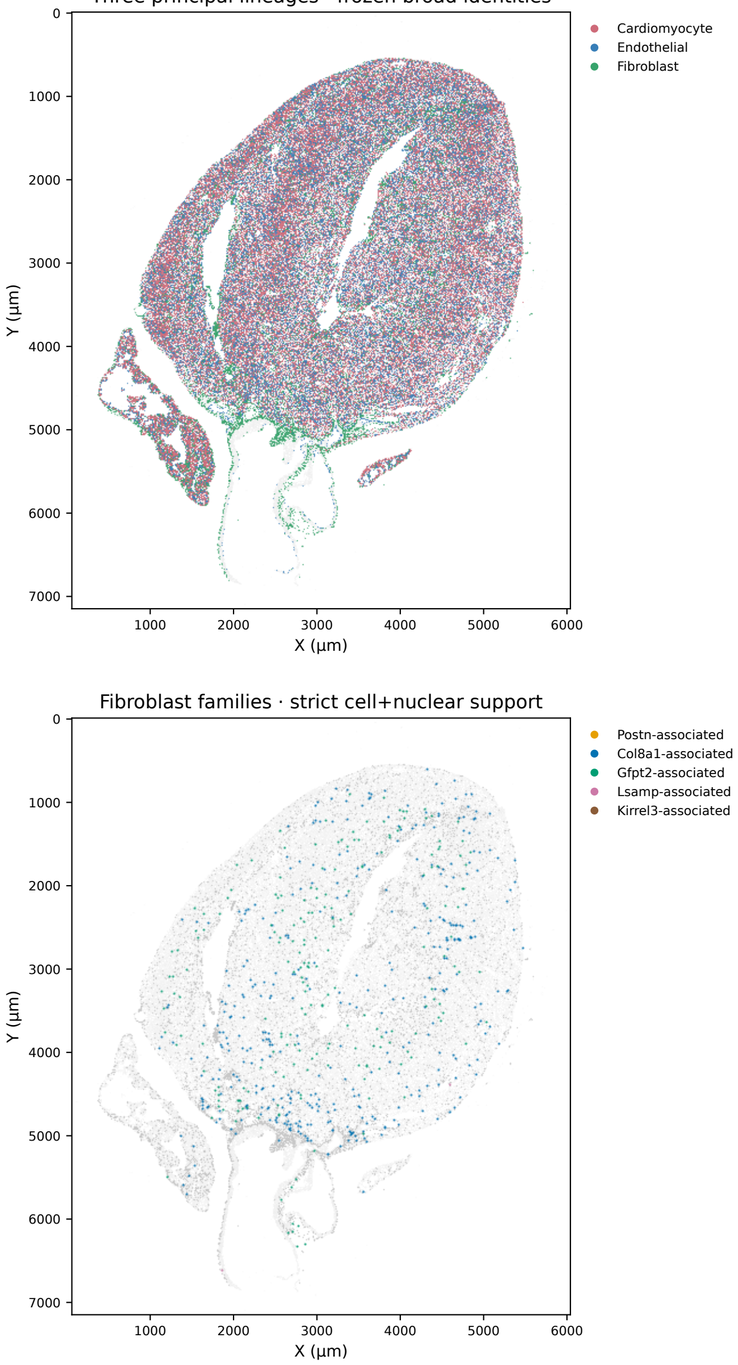


P7 · cardiac lineages, supported subtypes and ventricular maturation
Native section coordinates; gray fine identities are retained; confidence-selected coverage is incomplete

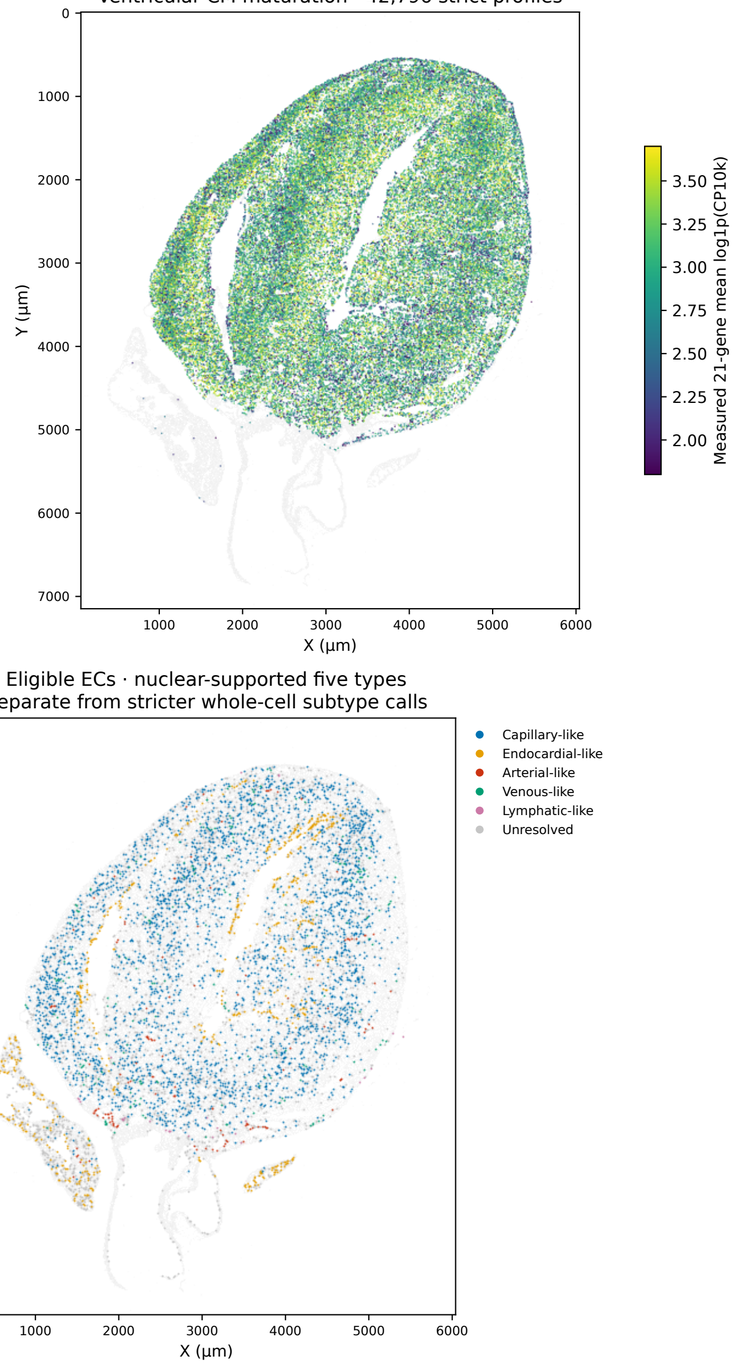


P14 · cardiac lineages, supported subtypes and ventricular maturation
Native section coordinates; gray fine identities are retained; confidence-selected coverage is incomplete

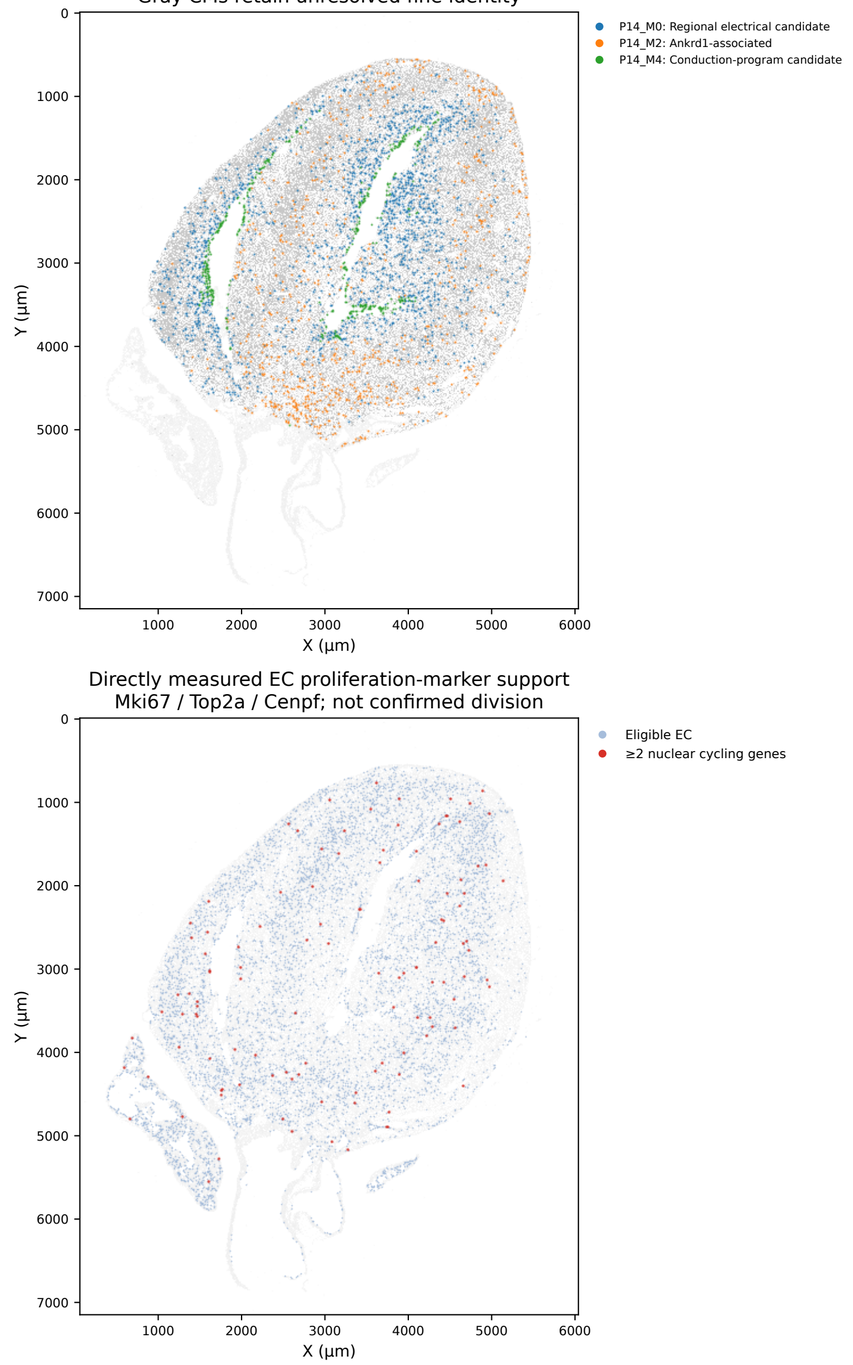
Three principal lineages · frozen broad identities



Ventricular CM maturation · 42,796 strict profiles

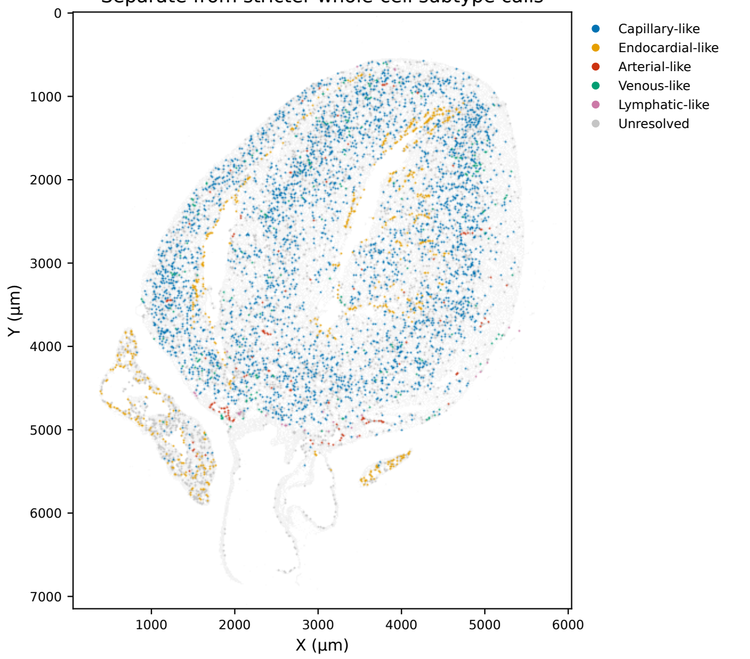


Prioritized fine ventricular candidates
Gray CMs retain unresolved fine identity



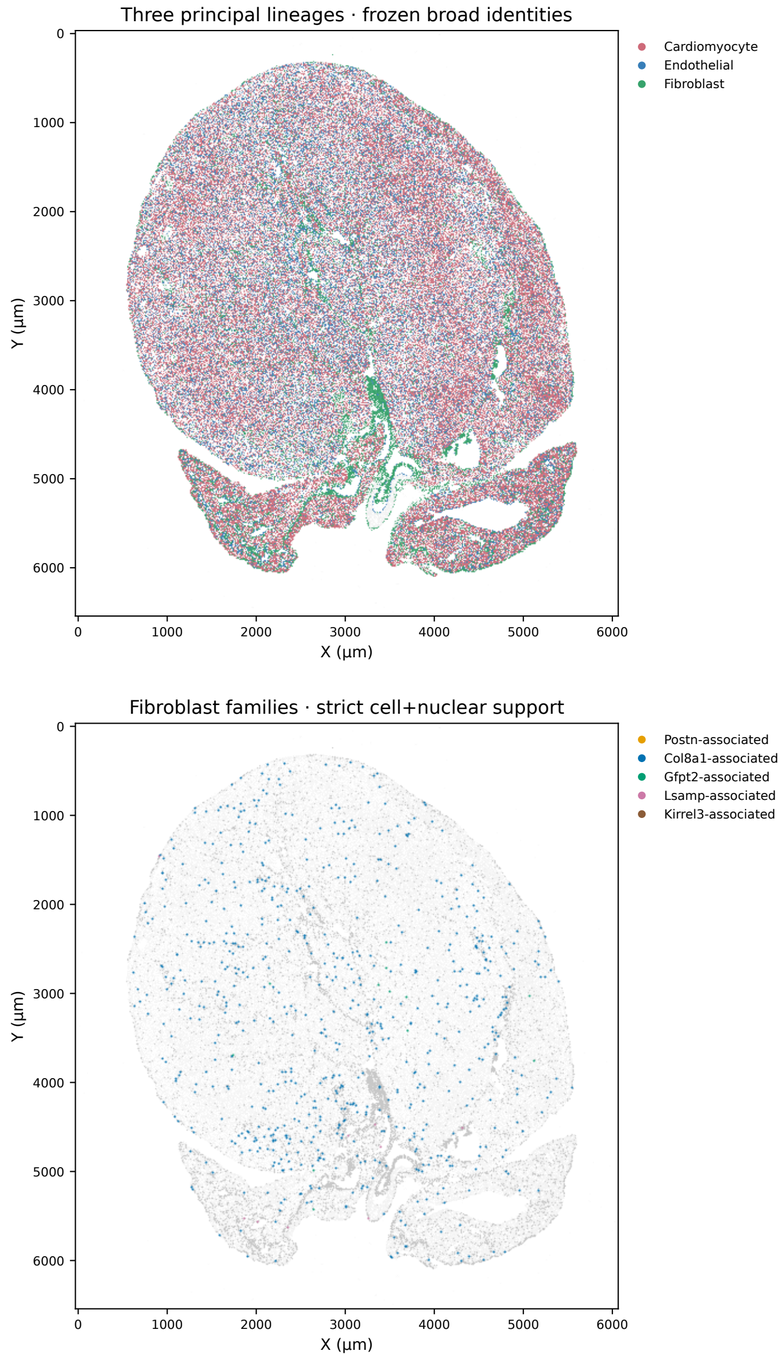
Fibroblast families · strict cell+nuclear support

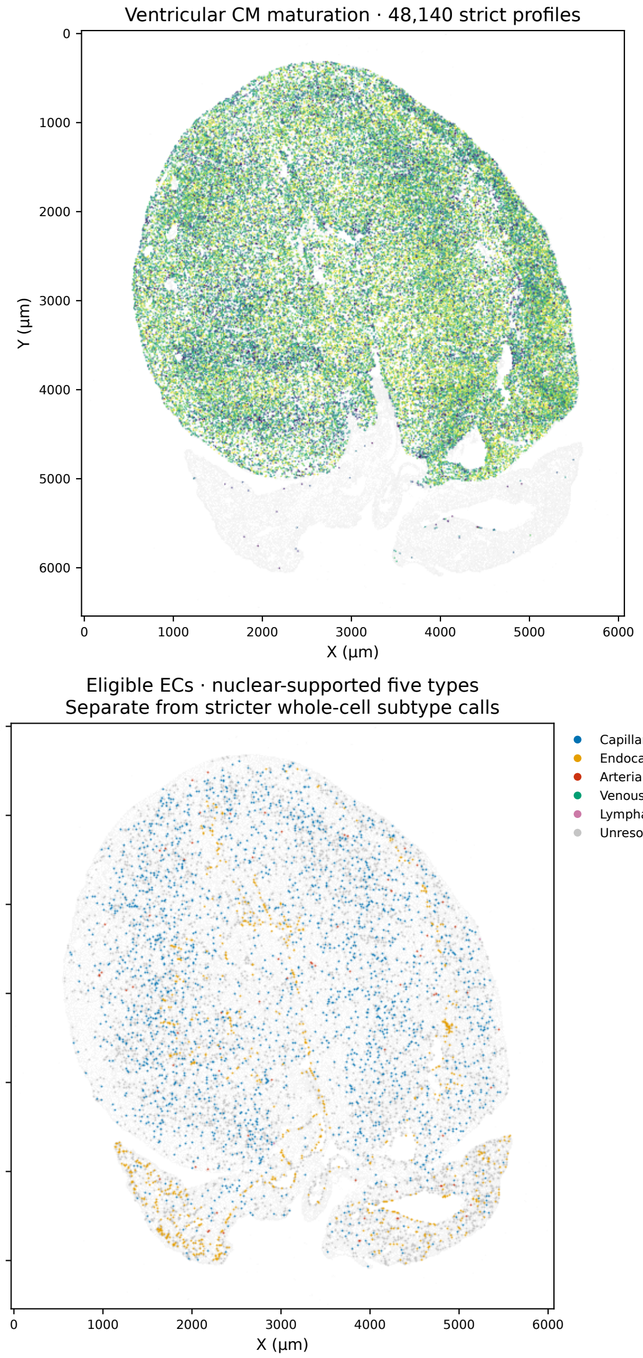
Eligible ECs · nuclear-supported five types
Separate from stricter whole-cell subtype calls

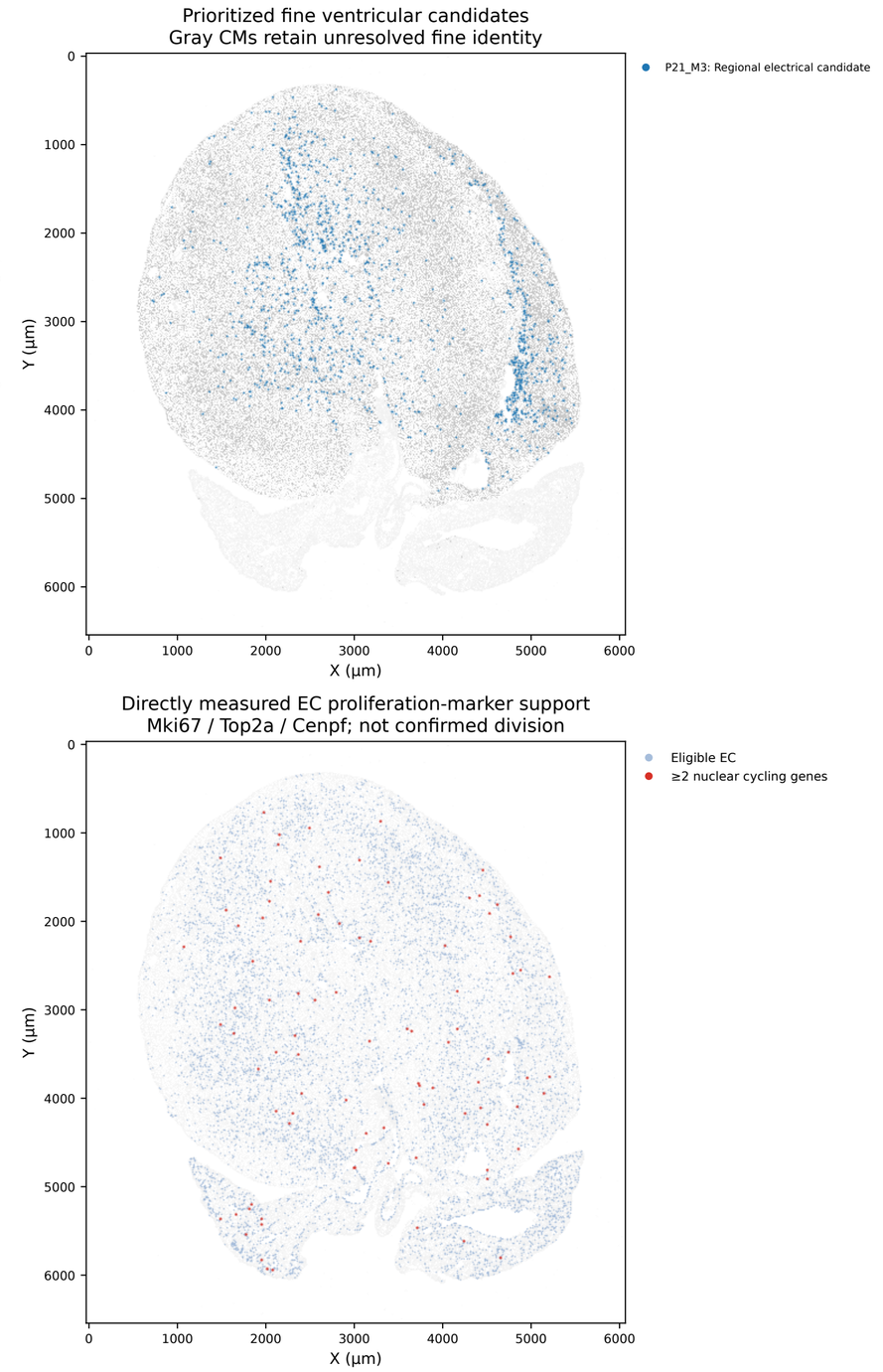


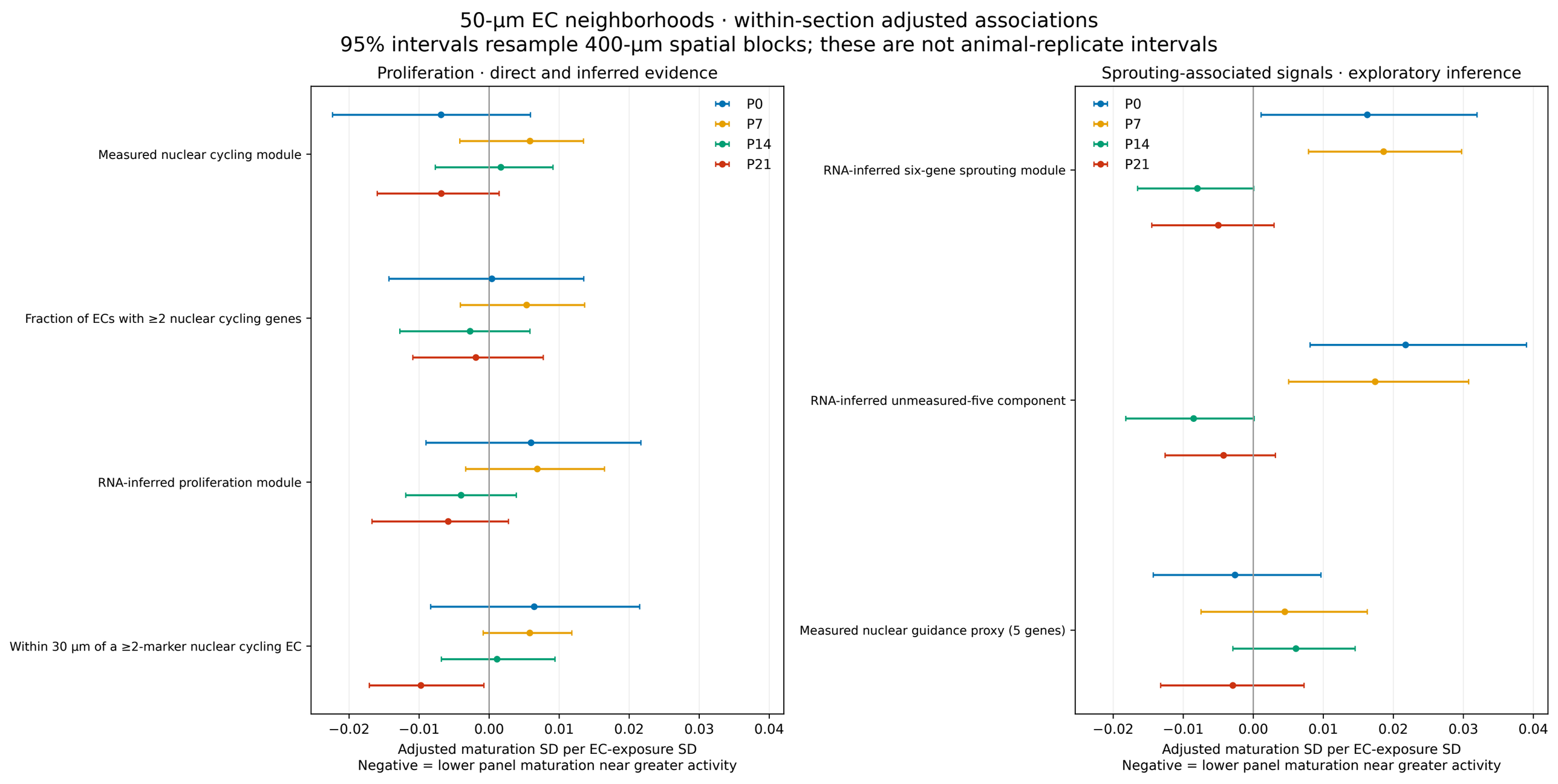
Directly measured EC proliferation-marker support
Mki67 / Top2a / Cenpf; not confirmed division

P21 · cardiac lineages, supported subtypes and ventricular maturation
Native section coordinates; gray fine identities are retained; confidence-selected coverage is incomplete

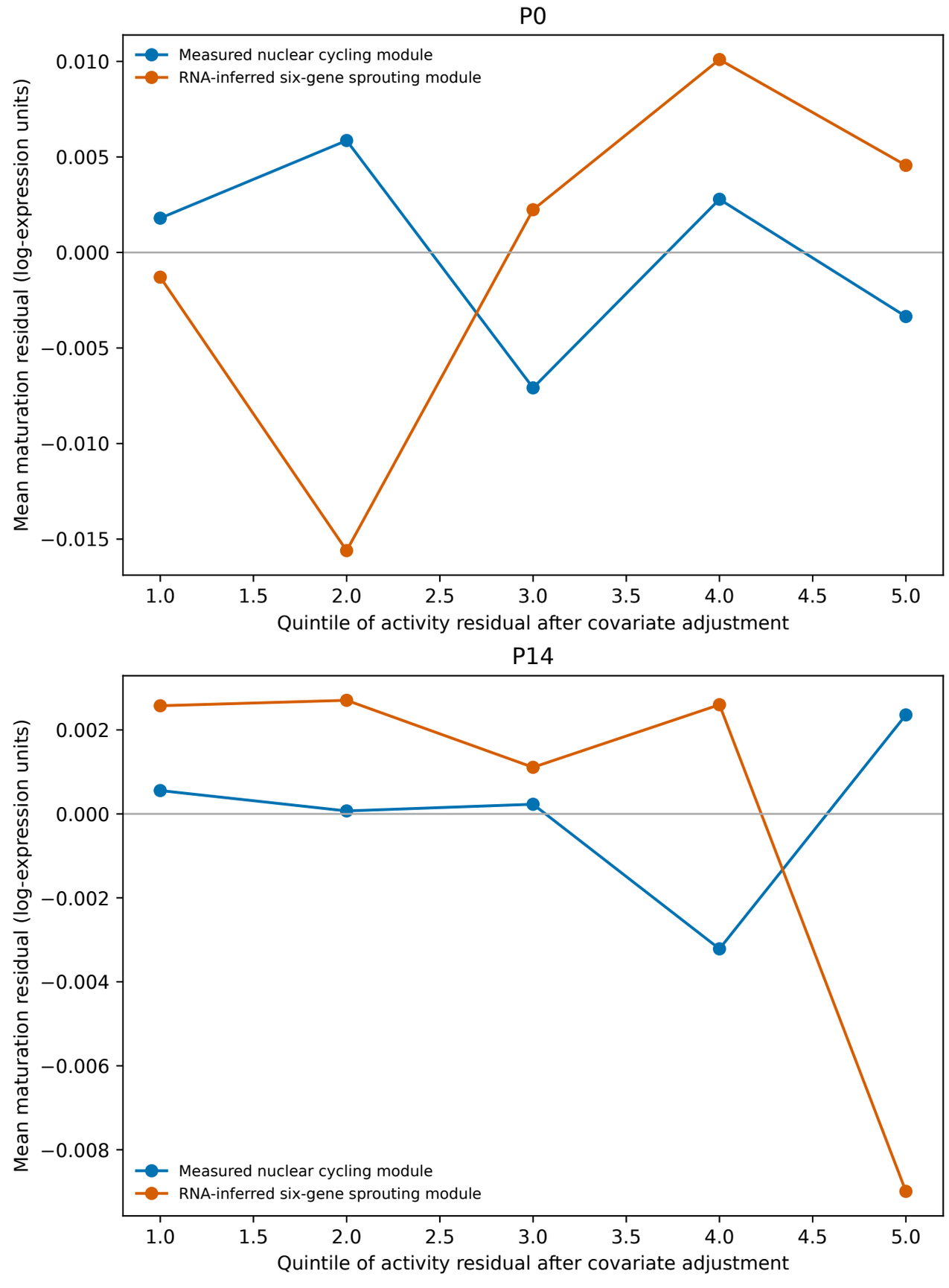


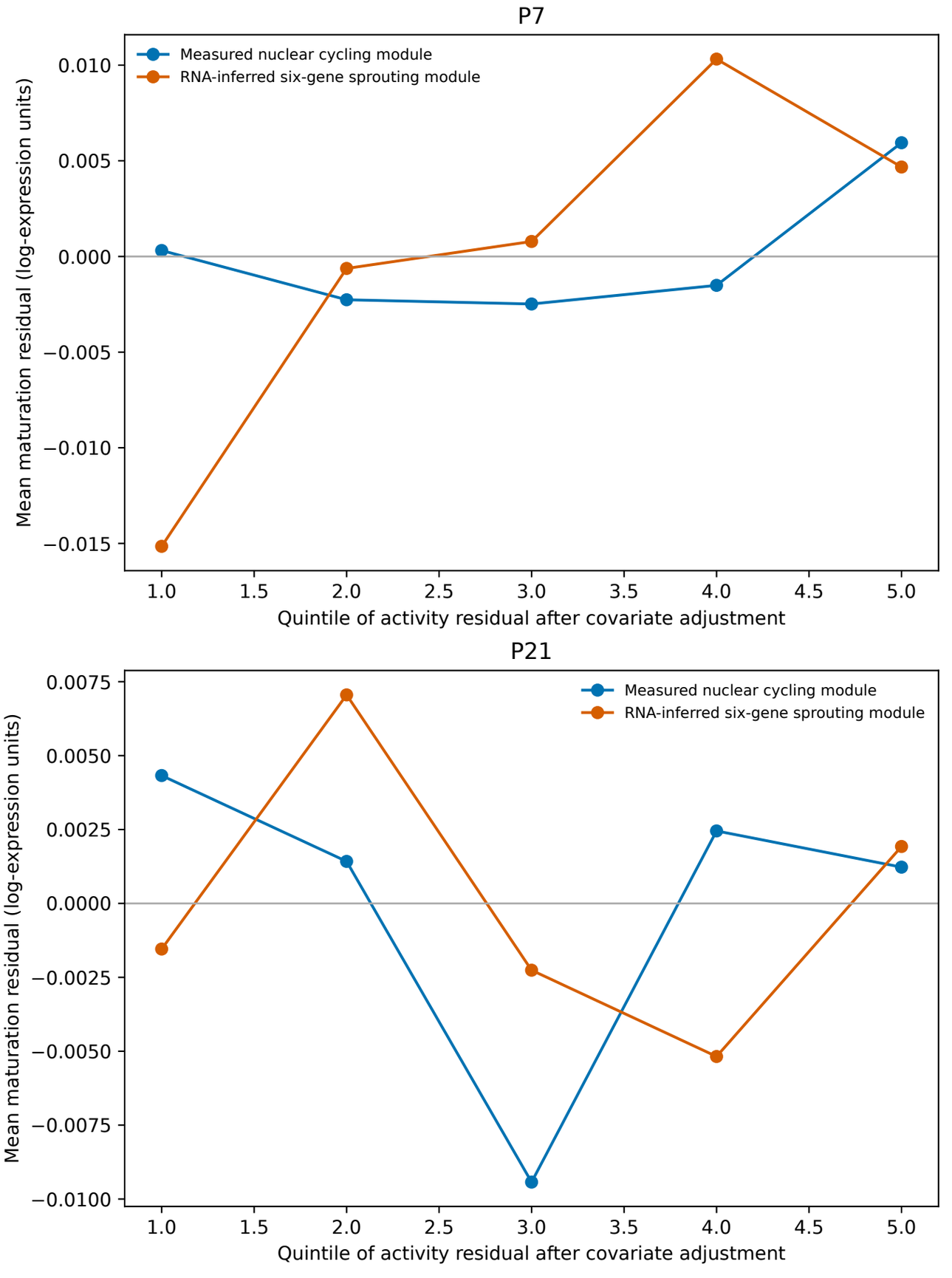




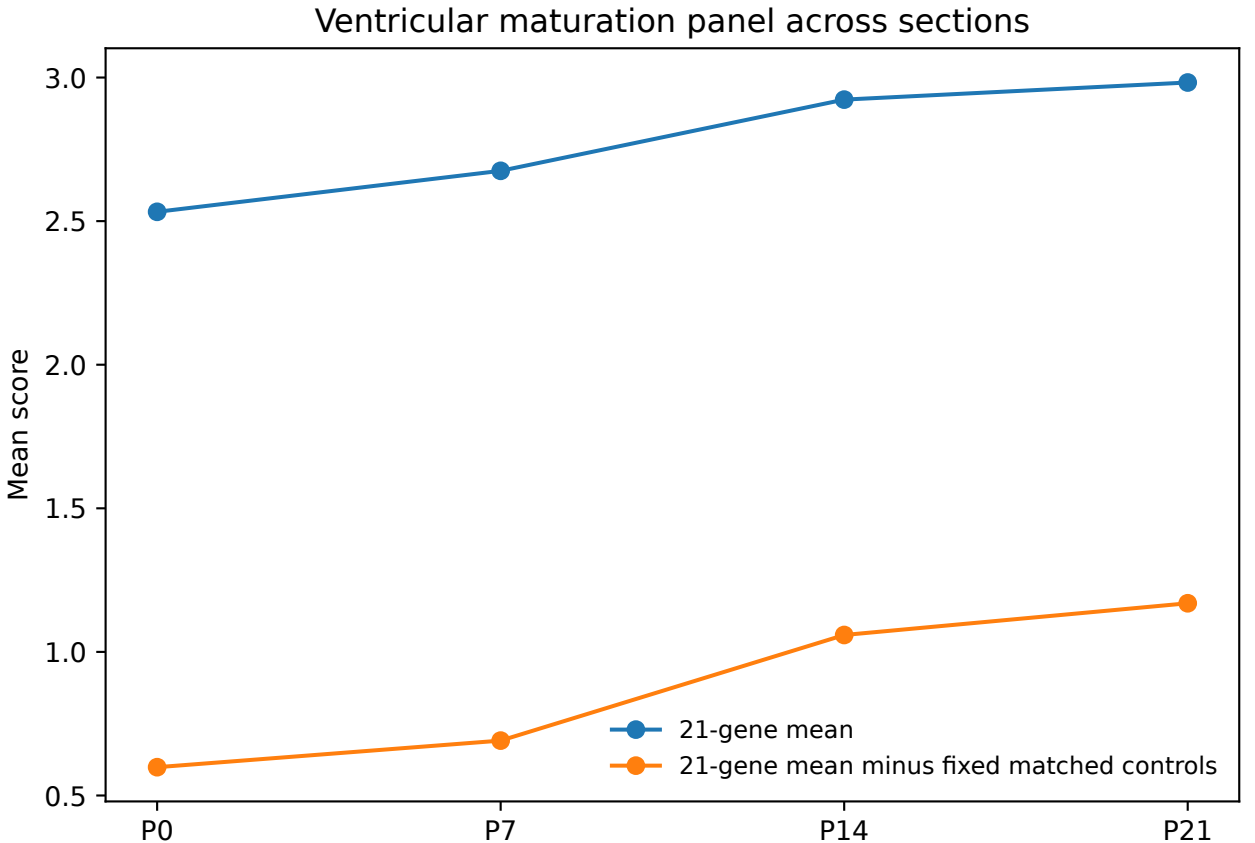

50-µm EC neighborhoods · within-section adjusted associations
95% intervals resample 400-µm spatial blocks; these are not animal-replicate intervals
Proliferation · direct and inferred evidence
Sprouting-associated signals · exploratory inference
P0
P7
P14
P21
Measured nuclear cycling module
Fraction of ECs with ≥2 nuclear cycling genes
RNA-inferred proliferation module
Within 30 µm of a ≥2-marker nuclear cycling EC
RNA-inferred six-gene sprouting module
RNA-inferred unmeasured-five component
Measured nuclear guidance proxy (5 genes)
−0.02
−0.01
0.00
0.01
0.02
0.03
0.04
Adjusted maturation SD per EC-exposure SD
Negative = lower panel maturation near greater activity

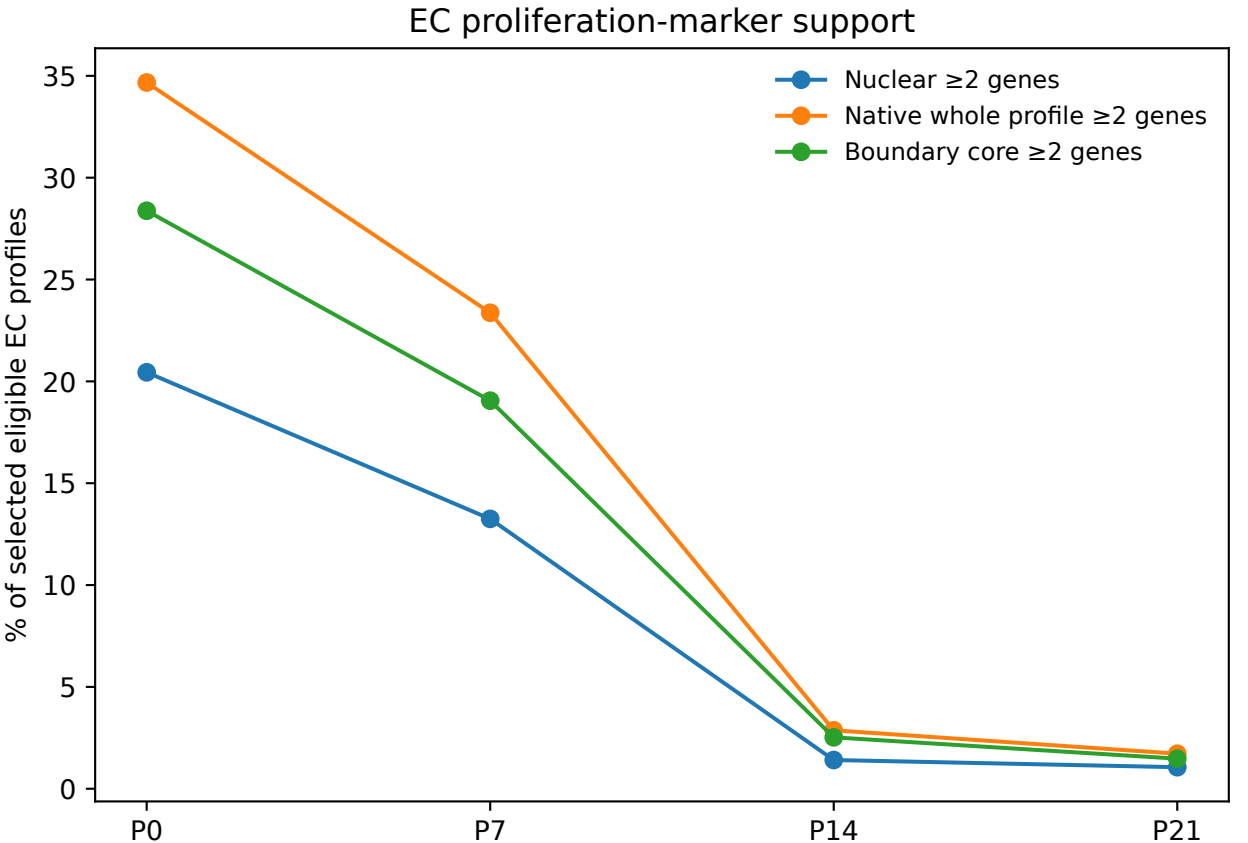
CM maturation across local EC activity gradients
Descriptive residual bins; overlapping neighborhoods are not independent replicates





Across-age trends and within-age proximity are different questions
One deposited section per age; capture and selection affect these descriptive fractions





Supported subtype neighborhoods · groups with ≥30 eligible profiles
EC strict and nuclear-supported are distinct evidence views, not additive populations

P0 Fibroblast Postn-associated
P0 EC strict Capillary-like
P0 EC strict Endocardial-like
P0 EC strict Arterial-like
P0 EC nuclear-supported Capillary-like
P0 EC nuclear-supported Endocardial-like
P0 EC nuclear-supported Arterial-like
P7 Fibroblast Gfpt2-associated
P7 Fibroblast Postn-associated
P7 EC strict Capillary-like
P7 EC strict Endocardial-like
P7 EC nuclear-supported Capillary-like
P7 EC nuclear-supported Endocardial-like
P14 Fibroblast Col8a1-associated
P14 Fibroblast Gfpt2-associated
P14 EC strict Capillary-like
P14 EC strict Endocardial-like
P14 EC strict Arterial-like
P14 EC strict Venous-like
P14 EC nuclear-supported Capillary-like
P14 EC nuclear-supported Endocardial-like
P14 EC nuclear-supported Arterial-like
P14 EC nuclear-supported Venous-like
P21 Fibroblast Col8a1-associated
P21 EC strict Capillary-like
P21 EC strict Endocardial-like
P21 EC strict Arterial-like
P21 EC nuclear-supported Capillary-like
P21 EC nuclear-supported Endocardial-like
P21 EC nuclear-supported Arterial-like

Cardiomyocyte
Endothelial
Fibroblast
Pericyte
Smooth muscle
Immune
Epicardial

Mean resolved-neighbor fraction (30 µm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6